Junu Lee

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Education

The Wharton School, University of Pennsylvania — Philadelphia, PA
Ph.D. in Statistics — 08/2022 – *Present*
Advisors: Zhimei Ren, T. Tony Cai

Harvard University — Cambridge, MA
A.B. in Mathematics and Statistics — 08/2018 – 05/2022
S.M. in Computer Science (concurrent AB/SM program)
Advisor: Lucas Janson

Research interests

Selective inference; Multiple hypothesis testing; Variable selection; Post-selection inference; e-values and e-processes; Sequential testing; Distribution-free inference;

Papers and preprints

** denotes alphabetical order or equal contribution*

[6] Abhinav Chakraborty*, **Junu Lee***, and Eugene Katsevich. "Power of masking methods for adaptive testing in a multivariate normal means problem". In: *arXiv preprint arXiv:2601.07764* (2026). Submitted to *Biometrika*.

[5] **Junu Lee**, Ilia Popov, and Zhimei Ren. "Full-conformal novelty detection: A powerful and non-random approach". In: *arXiv preprint arXiv:2501.02703* (2025). Major revision requested at *Journal of the American Statistical Association*.

[4] Jeffrey Zhang and **Junu Lee**. "A general condition for bias attenuation by a non-differentially mismeasured confounder". In: *Biometrika* 112.3 (2025), asaf026.

[3] **Junu Lee** and Zhimei Ren. "Boosting e-BH via conditional calibration". In: *arXiv preprint arXiv:2404.17562* (2024). Major revision requested at *Journal of the Royal Statistical Society, Series B*.

[2] Taehyeon Kim, Eric Lin, **Junu Lee**, Christian Lau, and Vaikkunth Mugunthan. "Navigating Data Heterogeneity in Federated Learning: A Semi-Supervised Approach for Object Detection". In: *Thirty-seventh Conference on Neural Information Processing Systems*. 2023.

[1] Eugene Curtin*, **Junu Lee***, Andrew Lu*, and Sophia Sun*. "A modified Grassmann algebra approach to theorems on permanents and determinants". In: *Linear Algebra and its Applications* 581 (2019), pp. 20–35.

Software

1. `fcnd`: Python package for full conformal novelty detection methods. [GitHub]

Awards and honors

IMS Hannan Graduate Student Travel Award — '26
Lawrence D. Brown Student Paper Award Symposium Speaker — '24, '25

NSF Graduate Research Fellowship	’22
Harvard College Research Program	’20
Herchel Smith Harvard Undergraduate Research Fellowship	’20
Derek Bok Certificate of Distinction in Teaching	’19, ’20

Talks

Invited talks:

8. **Power of masking methods for adaptive testing in a multivariate normal means problem**
   International Seminar on Selective Inference
   Virtual, April 2026
7. **Boosting e-BH via conditional calibration**
   Emmanuel Candès’ Group Meeting
   Stanford University, March 2026
6. **Full-conformal outlier detection**
   Lawrence D. Brown Student Paper Award Symposium
   University of Pennsylvania, October 2025
5. **Full-conformal outlier detection**
   Multiple Comparison Procedures (MCP)
   Philadelphia, August 2025
4. **Boosting e-BH via conditional calibration**
   BIRS Invited Workshop: Game-theoretic statistical inference
   Chennai, June 2025
3. **Using e-values for conformal multiple testing**
   Wharton Statistics First Year Ph.D. Student Seminar
   University of Pennsylvania, April 2025
2. **Boosting e-BH via conditional calibration**
   Lawrence D. Brown Student Paper Award Symposium
   University of Pennsylvania, November 2024
1. **Boosting e-BH via conditional calibration**
   International Seminar on Selective Inference
   Virtual, June 2024

Contributed talks:

3. **Boosting e-BH via conditional calibration**
   Joint Statistical Meetings (JSM)
   Nashville, August 2025
2. **Boosting e-BH via conditional calibration**
   New England Statistical Symposium (NESS)
   New Haven (given virtually), June 2025
1. **Boosting e-BH via conditional calibration (with applications to variable selection)**
   Eastern North American Region (ENAR)
   New Orleans, March 2025

Teaching

Teaching Assistant, The Wharton School

STAT 9270*‡: Bayesian Statistical Theory and Methods — Spring '24
STAT 4050‡/7050†‡: Statistical Computing with R — Fall '23 (Q2)
STAT 1110: Introductory Statistics — Fall '23
STAT 1020: Introductory Business Statistics — Fall '22

Teaching Assistant, Harvard University
STAT 149: Introduction to Generalized Linear Models — Spring '22
STAT 210*: Probability I — Fall '21
STAT 111: Introduction to Statistical Inference — Spring '20, '21
STAT 110: Introduction to Probability — Fall '19, '20, '21

**: graduate-level courses; †: MBA courses; ‡: guest lectures given*

Training

Undergraduate Student Research Supervision
– Ilia Popov (2024-2025).

Professional service

Wharton Directed Reading Program — January 2026 – *Present*
Organized and designed a directed reading curriculum on multiple hypothesis testing to undergraduate students.

Wharton Doctoral Program Ph.D. Mentor — August 2024 – *Present*
Provided mentorship to incoming Ph.D. students in the Wharton doctoral programs.

Invited Discussion
International Seminar on Selective Inference:
– "E-statistics, group invariance and anytime-valid testing" by Muriel Pérez-Ortiz, December 2024
– "Bringing Closure to FDR Control With a Uniform Improvement of the e-Benjamini-Hochberg Procedure" by Neil Xu, May 2025

Conferences
– Chair of the invited session "Conformal inference and statistical testing for reliable deployment of AI/ML models" at Joint Statistical Meetings (JSM), 2025
– Organizer and co-chair of the invited session "The role of e-values in multiple testing" at Multiple Comparison Procedures (MCP), 2025

Journal Reviewer: Journal of the American Statistical Association (1)

Industry experience

Dynamo AI — *Various*
Founding member, advisor — 05/2022 – 12/2023
– Developed a multitude of federated learning algorithms and adapted them to various problem domains, e.g., finance, automated driving.
– Met with industry partners to collaborate on algorithms and present finished products.

Hudson River Trading New York, NY

Algorithm developer intern 06/2021 – 08/2021

– Completed multiple quantitative projects on financial forecasting.

Proton.ai Boston, MA

ML research intern 05/2019 – 08/2019

– Developed NLP-based recommendation systems for sales teams to efficiently market new products and promotions.

– Contributed to the research and training of customer churn models.